Barış Salih Babacan

Systems Engineer — Rust · Go · Swift

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SUMMARY

Systems engineer building local-first infrastructure in Rust, with production experience spanning embedded systems, network protocol reverse engineering, and native applications. Shipped Sortify, an iOS/iPadOS game-based learning platform now at version 2.3 on the App Store; reverse-engineered Philips' undocumented JointSpace API for +TV, a shipped Android TV remote with 10K+ downloads; and delivered a hardware access-control system that served 120–150 users a day for three academic years.

SKILLS

- **Systems & Protocols:** Rust, content-addressable storage (BLAKE3), Unix domain sockets, JSON-RPC, IPC & sidecar processes, HTTP Digest auth, mDNS/DIAL discovery, serial & embedded (Arduino), Git internals
- **Languages:** Rust, Swift, Go, TypeScript, JavaScript, C++, Python, SQL
- **Apple & Desktop:** SwiftUI, MVVM, Swift Package Manager, Tauri, App Store release
- **Backend & Web:** Node.js, Express, Go (net/http), React, Next.js, Firebase (Firestore, Functions, Auth), MongoDB, SQLite
- **Practice & Tooling:** Git, GitHub Actions CI, code review workflows, unit testing & benchmarking, Wireshark, Linux

EXPERIENCE

CTO & Lead Developer | *Garage.ist* | Oct 2024 – Present
Product studio building consumer iOS and Android applications.

- Lead development of **Sortify**, a game-based learning app for ages 3–13 with parent and teacher supervision tools (SwiftUI + MVVM + Firebase), now at **v2.3 on the App Store**; I wrote **450+ of its 550+ commits**.
- Split the codebase into six in-house Swift packages for core, auth, education, games, subscription and chessboard; integrated subscriptions, push notifications and analytics.
- Led development with a two-developer team and enforced review discipline through CI that blocks direct pushes to main and self-merged pull requests.
- Co-built **+TV**, a shipped Android TV remote, with a fellow developer, integrating Samsung through the Tizen SDK and LG through ConnectSDK.
- Wrote the Philips integration myself where no SDK existed, reverse-engineering the undocumented JointSpace API from Philips' own remote app (logs via CatLog, traffic via Wireshark): HTTP Digest authentication over a 401 challenge–response flow, PIN-based device pairing as an explicit state machine, and TLS handling for self-signed certificates.
- Developed three shipped iOS games in Swift and SwiftUI: **Guess the City** (daily geography puzzle integrating the WorldClock 1,000+ city dataset with offline Haversine distance/direction math), **Block Nest** (custom gameplay mechanics and rotation modes), and **Rule Sort** (foundation, Firebase, and AdMob integration).

IT & Systems Integration Intern | *ENLOG, Istanbul* | Sep 2023 – Jun 2024

- Supported network operations, technical support and internal IT infrastructure while completing high school.

PROJECTS

- **Chorus — anonymous discussion engine** | *Go, React, Git* | github.com/joinchorus/chorus
  Built and maintain an open-source forum that stores every thread and message as an immutable Git commit, with no database in the dependency tree. Thread-scoped ephemeral identities replace user accounts. Shipped v0.1.0-alpha with IP token-bucket rate limiting tuned per endpoint, X-Request-ID correlation across logs and error payloads, and a 10-second graceful shutdown that never interrupts an in-flight Git write. 24 tests and 5 benchmarks measuring the cost of Git-as-storage.
- **Osmos — local-first version control engine** | *Rust, Tauri, React* | github.com/Osmos-App/osmos-core
  Designed a three-crate Rust workspace: content-addressable object store keyed by BLAKE3 with streaming file hashing, a SQLite-backed tree index, and a Unix-socket daemon exposing ten typed commands, among them init, status, branch, switch, merge and commit, over a UUID-correlated request/response protocol. **Zero unsafe blocks.** Early-stage Tauri/ React client; peer-to-peer sync is planned, with the transport crate scaffolded and not yet implemented.

- **Mythos — local-first IDE for novelists** | *Rust, Tauri, TypeScript* | github.com/Mythos-IDE/mythoside-core
  Built a standalone Rust engine across six crates that models manuscripts as Markdown + YAML on disk and exposes 13 JSON-RPC methods over stdio, run by a Tauri/React client as a managed sidecar. Types are generated from Rust into TypeScript so the IPC boundary stays type-safe. **82 tests.**
- **Q-PASS — RFID access control** | *C++, Arduino, Python, PHP, MySQL*
  Delivered an end-to-end hardware and software system for my vocational high school: C++ microcontroller firmware bridged to a web backend over serial, in custom 3D-printed reader enclosures. **Served 120–150 people daily for three academic years**, handling cafeteria access and meal tracking.
- **IT-RMS — IT resource management** | *PHP, MySQL, Node.js, Express, MongoDB*
  Built the IT asset system for my vocational high school's IT department across 249 commits, starting in 2023 during the third year of high school: multi-branch inventory with barcode generation, PDF reporting, audit logging and a five-role user model. Returned to it in 2026 and am **migrating the production codebase from PHP/MySQL to Express 5 + MongoDB**, keeping the legacy version on a branch and restructuring it into an MVC layout with linting, input validation and security headers.

EDUCATION

Istanbul University (Open & Distance Education Faculty) | *Associate Degree, Computer Programming* | 2024 – Present

Haydarpaşa Vocational & Technical High School | *Network Management & Cyber Security* | 2020 – 2024